

SmartPrint Kiosk

Revolutionizing Campus Printing at VIT Chennai

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The Campus Printing Bottleneck

- **Severe Queueing:** During assignment submissions, students waste critical academic hours waiting at crowded stationery shops.
- **Staff Dependency:** Print access is strictly limited by shop operating hours and the availability of personnel.
- **Transaction Friction:** Dealing with exact cash or finding change for minimal amounts (₹2 to ₹10) causes severe delays.
- **Privacy & Security Risks:** Forwarding personal or sensitive assignments to shared shop computers via WhatsApp or email exposes data.



The SmartPrint Solution

100% Automated. Zero Hassle.

We are proposing a fully automated, self-service printing kiosk network tailored specifically for the Indian campus ecosystem.

By merging a custom Python-based print spooler with an intuitive tablet interface and instant UPI payments, we eradicate queues, manual dependencies, and cash handling entirely.



Seamless 3-Step User Journey



1. Initiate

Students send documents to the virtual "SmartPrint" printer from their laptop. A prompt collects their name and preferences (Color/BW, Copies).



2. Authenticate

At the Xerox machine, the student finds their name on the attached tablet's digital queue and verifies the dynamically calculated cost.



3. Pay & Print

A dynamic UPI QR code appears on the tablet. The student scans, pays, and upon instant confirmation, the hardware immediately executes the print.

System Architecture

The Python Virtual Spooler

A custom-built, lightweight Python software acts as a virtual printer on the student's device. Instead of printing immediately, it halts the job, attaches essential metadata (Name, Print Settings), and securely transmits the payload to our central cloud-based print queue.

The Kiosk Command Center

A dedicated application runs on a tablet physically mounted to the Xerox machine. It fetches the live queue, generates dynamic UPI payment links via API, and commands the printer hardware to release the job only upon receiving a successful webhook payment receipt.

The Power of UPI Integration

100%

Cashless Efficiency

Built for Digital India

While Western universities rely on clunky prepaid ID cards, proprietary top-up portals, or coin slots, our system leverages India's globally leading UPI infrastructure.

This means absolutely **zero onboarding friction**. If a student has a smartphone, they can print. No account balances to manage, and total transparency on every transaction.

Global Model, Local Innovation

Self-service campus utilities are a global university standard. We aren't reinventing the wheel; we are perfecting it for the Indian ecosystem with instant UPI architecture.

— The SmartPrint Vision for VIT

Strategic Advantages for VIT

-  **24/7 Availability:** Kiosks can be placed directly in hostel blocks, enabling students to print reports and assignments at 2 AM without ever leaving their buildings.
-  **Reduced Operational Overhead:** Entirely eliminates the need for dedicated manpower to monitor printers, collect cash, and actively manage long queues.
-  **Enhanced Data Security:** Documents are routed directly to the printer buffer and purged post-printing, protecting student intellectual property and privacy.
-  **Centralized Analytics:** University management gains access to a robust dashboard detailing usage metrics, peak print times, and digital revenue generation.

The Technology Stack

 Python code

Client App: Python-based print interceptor and dynamic UI prompt.

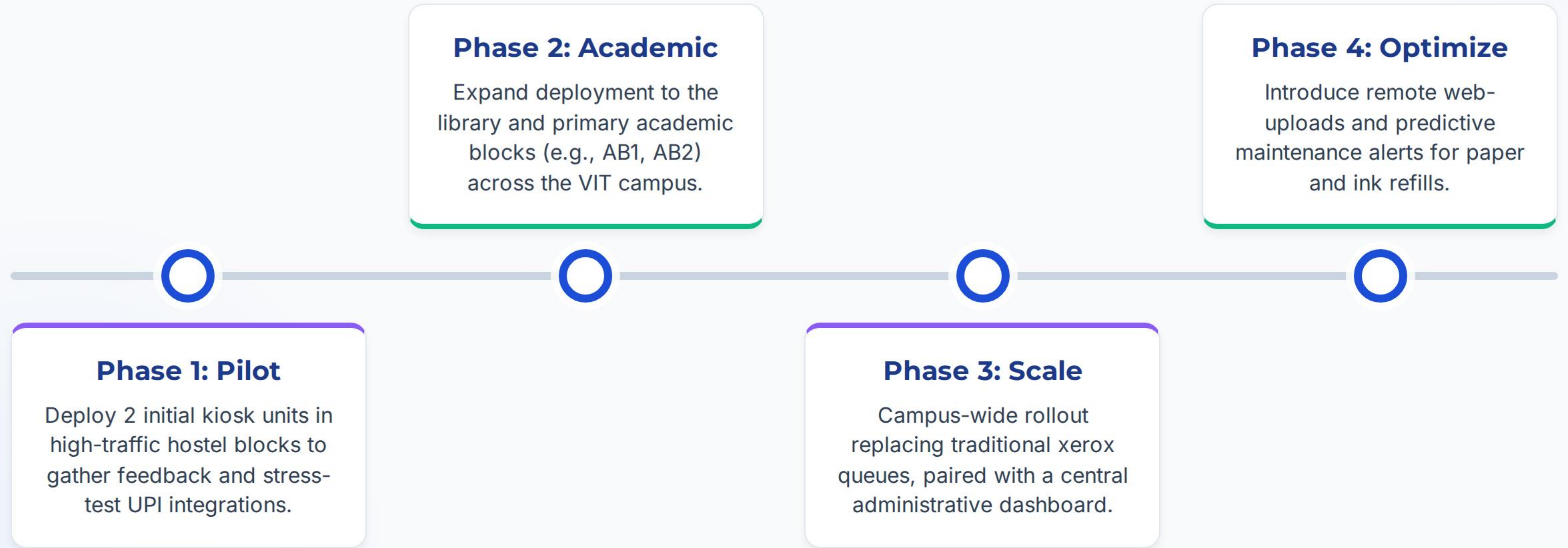


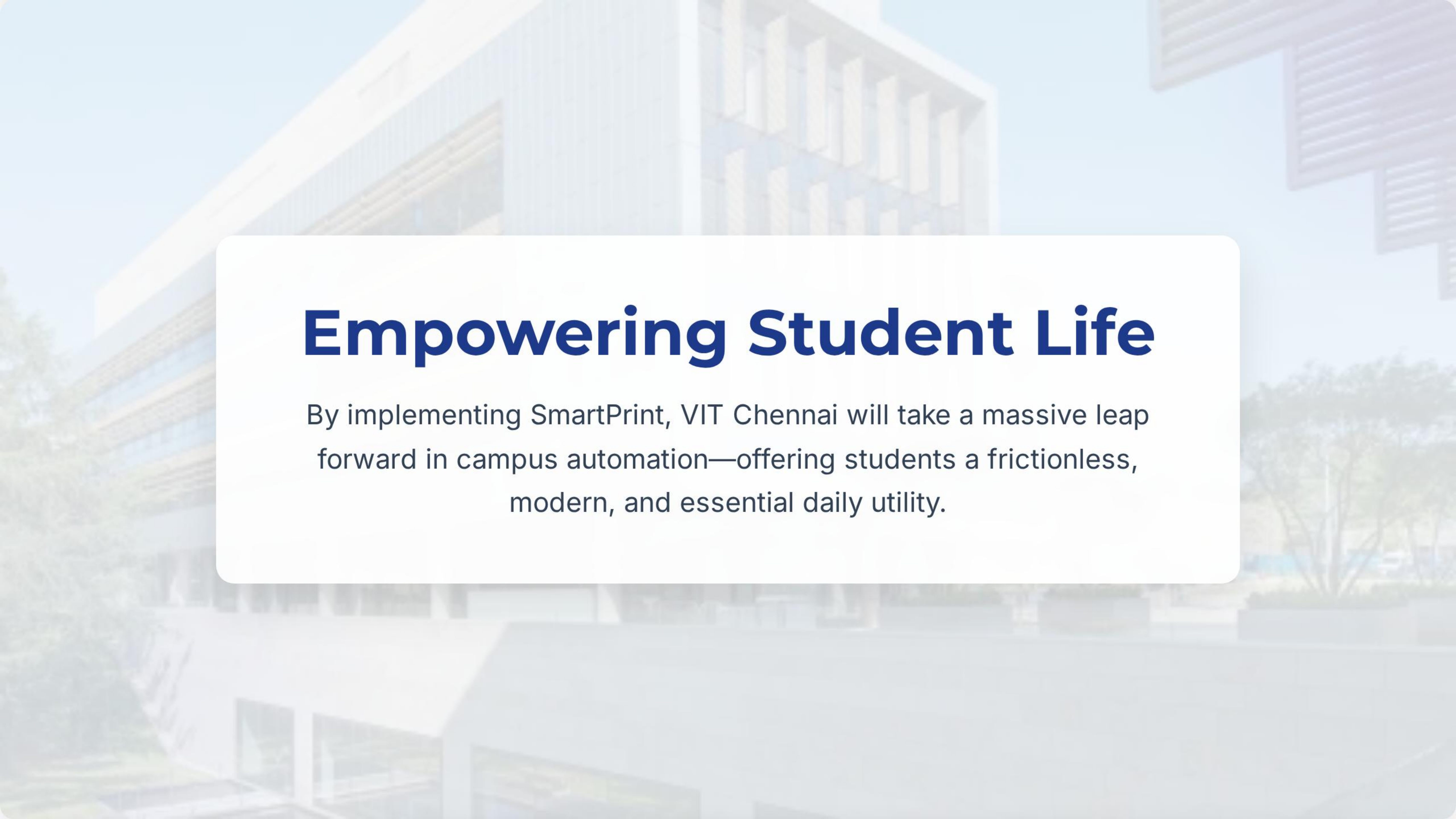
Kiosk Interface: Interactive tablet application managing the job queue.



Payment Gateway: Dynamic UPI QR generation & real-time webhooks.

Implementation Roadmap





Empowering Student Life

By implementing SmartPrint, VIT Chennai will take a massive leap forward in campus automation—offering students a frictionless, modern, and essential daily utility.

Ready to Innovate?

Thank you for your time and consideration.

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Image Sources



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